

How long is the graph of x^n ?

Logarithmic corner defects in arc length

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Abstract

The graph of $y = x^n$ on $[0, 1]$ has arc length $L(n)$ that tends to 2 as $n \rightarrow \infty$, because the curve hugs the two legs of the corner $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1)$. We determine exactly how fast. We prove

$$2 - L(n) = \frac{\ln n + 1 - \ln 2}{n - 1} + O\left(\frac{\ln^2 n}{n^2}\right),$$
$$L(n) = 1 + \left(\frac{2}{e}\right)^{1/(n-1)} + \frac{\pi^2}{12n^2} + O\left(\frac{\ln n}{n^3}\right).$$

Both statements follow from an exact, rapidly convergent formula for $L(n)$ involving only the Gamma function, which we obtain from a single Beta integral. The factor $\ln n$ has a simple geometric cause: x^n spends a horizontal distance of about $(\ln n)/n$ climbing steeply. The phenomenon persists for corners of any turning angle θ : the coefficient of $(\ln n)/n$ becomes $2\sin^2(\theta/2)$, and we compute the next constant in closed form. For contrast, we show that the perimeter of the “squircle” $|x|^p + |y|^p = 1$ approaches 8 *without* a logarithm: $P(p) = 8 - 4K/p + O(p^{-2})$ with $K = 2\sqrt{2} \ln(1 + \sqrt{2}) - 2\ln 2$. All results are elementary and accessible to a reader who knows calculus and the Gamma function.

Keywords: arc length, asymptotic expansion, Mellin transform, Beta integral, superellipse.

MSC 2020: 26A06, 41A60, 33B15, 53A04.

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1 Introduction

A classic warning in calculus concerns the “staircase paradox”: a staircase from $(0,0)$ to $(1,1)$ with ever finer steps converges uniformly to the diagonal, yet its length stays equal to 2 rather than approaching $\sqrt{2}$. Arc length is not continuous under uniform convergence; it is only lower semicontinuous [1].

The family $y = x^n$ on $[0,1]$ exhibits a similar effect, in a smooth and more subtle form. As $n \rightarrow \infty$ the graphs converge (in the Hausdorff sense) to the “corner” path

$$\Gamma : (0,0) \longrightarrow (1,0) \longrightarrow (1,1),$$

and this time the lengths do converge to the length of the limit:

$$L(n) := \int_0^1 \sqrt{1 + n^2 x^{2n-2}} dx \longrightarrow 2.$$

This is a common exercise. The question we answer here is: *how fast?* The answer turns out to be surprisingly rich (see Figure 1).

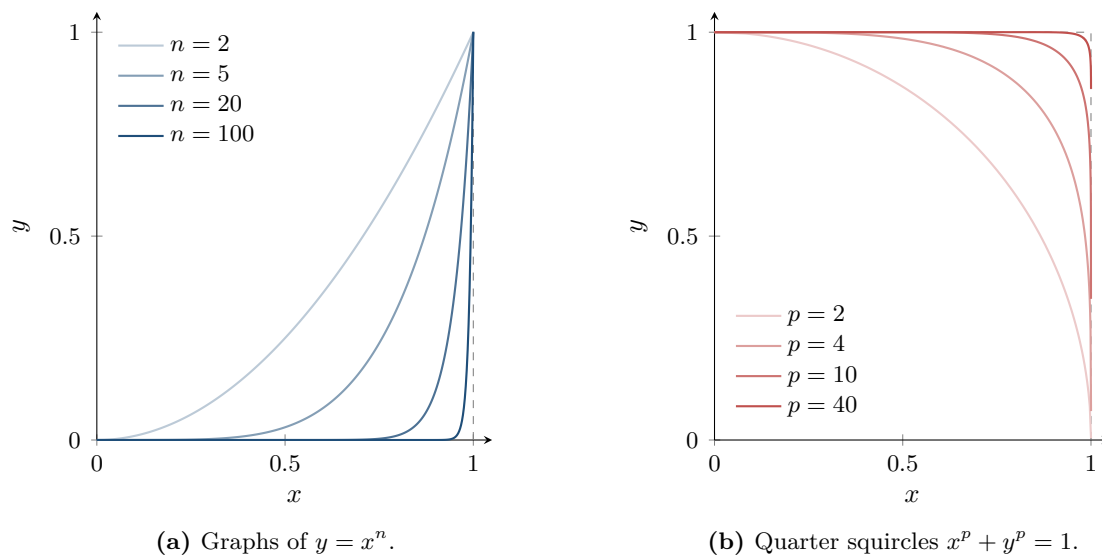


Figure 1: Two families of curves that collapse onto a right-angled corner (dashed). In (a) the length defect is of order $(\ln n)/n$ (Theorems A and B). In (b) it is of order $1/p$ (Theorem D).

1.1 Main results

Throughout, $n > 1$ is real, $\ln$ is the natural logarithm, and we write

$$\varepsilon := \frac{1}{n-1}, \quad D(n) := 2 - L(n).$$

Our first result is an elementary estimate, with explicit constants.

Theorem A (Proposition 3.1). *For every $n \geq 3$,*

$$\frac{\ln n - 1}{n - 1} - \frac{\ln^2 n}{2(n - 1)^2} \leq 2 - L(n) \leq \frac{\ln n + 1}{n - 1}.$$

In particular $2 - L(n) \sim (\ln n)/n$.

The second result is exact. Let

$$R(\varepsilon) := \frac{\Gamma(1 + \frac{\varepsilon}{2}) \Gamma(\frac{1-\varepsilon}{2})}{\sqrt{\pi} (1 + \varepsilon)}, \quad 0 \leq \varepsilon < 1, \quad (1.1)$$

so that $R(0) = 1$.

Theorem B (Theorem 4.2, Corollary 5.2). *For every real $n > 2$, with $\varepsilon = 1/(n-1)$,*

$$L(n) = 1 + \frac{R(\varepsilon)}{n^\varepsilon} - \sum_{k=1}^{\infty} \binom{1/2}{k} \frac{1}{n^{2k-1}((2k-1)(n-1)-1)}. \quad (1.2)$$

Consequently

$$L(n) = 2 - \frac{\ln n + 1 - \ln 2}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right), \quad (1.3)$$

$$L(n) = 1 + \left(\frac{2}{e n}\right)^{1/(n-1)} + \frac{\pi^2}{12 n^2} + O\left(\frac{\ln n}{n^3}\right). \quad (1.4)$$

The terms of the series in (1.2) decay like n^{-2k} . Four terms already give $L(100)$ to about 22 digits. Formula (1.4) is remarkably accurate: at $n = 100$ its error is below 5×10^{-6} (Table 2).

Where does $\ln n$ come from? Section 8 explains that it is a feature of how x^n approaches the *vertical* leg of the corner. Section 6 shows that it does not depend on the corner being a right angle. For unit vectors u, v making an angle $\theta \in [0, \pi)$, consider the curve $\gamma_n(x) = x u + x^n v$, $0 \leq x \leq 1$, which collapses onto the two-segment path with legs u and v (total length 2) as $n \rightarrow \infty$. The graph of x^n is the case $\theta = \pi/2$.

Theorem C (Theorem 6.2). *Let $L_\theta(n)$ be the length of γ_n . Then*

$$2 - L_\theta(n) = \frac{2 \sin^2(\theta/2) \ln n + C(\theta)}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right), \quad C(\theta) = 2 \sin^2 \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2} \ln \cos \frac{\theta}{2}.$$

Finally, the logarithm is *not* a universal feature of curves collapsing onto a corner. The superellipse (or “squircle”) $|x|^p + |y|^p = 1$, popularized by Piet Hein [4], converges to the square $[-1, 1]^2$ as $p \rightarrow \infty$, but more quickly.

Theorem D (Theorem 7.1). *The perimeter $P(p)$ of $|x|^p + |y|^p = 1$ satisfies, for $p \geq 2$,*

$$\left| P(p) - \left(8 - \frac{4K}{p}\right) \right| \leq \frac{12}{p^2}, \quad K = 2\sqrt{2} \ln(1 + \sqrt{2}) - 2 \ln 2 = 1.10660 \dots$$

Table 1 collects these results.

1.2 Related work and what is new

The limit $L(n) \rightarrow 2$ is standard. The indefinite arc-length integral of x^n is known to be a Gauss hypergeometric function; see the discussion in [5], which gives in particular

$$L(n) = {}_2F_1\left(-\frac{1}{2}, \frac{\varepsilon}{2}; 1 + \frac{\varepsilon}{2}; -n^2\right). \quad (1.5)$$

In principle, (1.2) can be obtained from (1.5) through the classical connection formula that transforms a ${}_2F_1$ at argument z into functions of $1/z$ [2, 15.8.2] (see Remark 4.5). Our derivation avoids hypergeometric functions entirely. It uses one substitution and one Beta

Table 1: Summary of the main results. In each case the curve collapses onto a two-segment path, and the *defect* is the limiting length minus the actual length.

Curve	Limit	Defect	Where
$y = x^n, 0 \leq x \leq 1$	2	$\frac{\ln n + 1 - \ln 2}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right)$	Thm. 4.2, Cor. 5.2
$xu + x^n v$, angle θ between u, v	2	$\frac{2 \sin^2 \frac{\theta}{2} \ln n + C(\theta)}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right)$	Thm. 6.2
squircle $ x ^p + y ^p = 1$ (perimeter)	8	$\frac{4K}{p} + O\left(\frac{1}{p^2}\right)$	Thm. 7.1

integral, and it leads directly to the asymptotic statements (1.3)–(1.4), which we have not found in the literature. Two recent preprints [6, 7] give hypergeometric series representations for the arc length of supercircles and the perimeter of Lamé superellipses, and they verify the limiting value 8 as $p \rightarrow \infty$. They do not give the rate $4K/p$ of Theorem D, and we did not find it elsewhere either. Given how elementary these questions are, some of our results may well exist as solved problems in the journal literature. We would welcome references.

1.3 Organization

Section 2 reduces every result to a single one-dimensional integral. Section 3 proves Theorem A, Section 4 proves the exact formula, and Section 5 extracts the asymptotics. Section 6 treats general angles, Section 7 treats the squircle, and Section 8 explains the difference between the two behaviors. Section 9 summarizes and lists open questions.

2 The taxicab defect

The key idea is to compare Euclidean length with *taxicab* length. Let $\gamma(t) = (x(t), y(t))$ be a C^1 curve whose two coordinates are each monotone, joining $(0, 0)$ to $(1, 1)$. Its taxicab length is $\int (|x'| + |y'|) dt = 1 + 1 = 2$. Hence

$$2 - \text{length}(\gamma) = \int \left(|x'| + |y'| - \sqrt{x'^2 + y'^2} \right) dt \geq 0. \quad (2.1)$$

The integrand is nonnegative. It vanishes where the curve is horizontal or vertical and is largest where the curve moves diagonally. Thus $2 - L(n)$ measures how much “diagonal travel” the curve does. The same idea works for any pair of directions.

Lemma 2.1 (Defect integral). *Let $u, v \in \mathbb{R}^2$ be unit vectors with $u \cdot v = c = \cos \theta$, where $\theta \in [0, \pi]$. Let $n > 1$, $\varepsilon = 1/(n-1)$, and $\gamma_n(x) = xu + x^n v$ for $0 \leq x \leq 1$. Then*

$$2 - \text{length}(\gamma_n) = \varepsilon n^{-\varepsilon} \int_0^n h_c(s) s^{\varepsilon-1} ds, \quad (2.2)$$

where

$$h_c(s) := 1 + s - \sqrt{1 + 2cs + s^2} = \frac{2(1-c)s}{1 + s + \sqrt{1 + 2cs + s^2}} \quad (s \geq 0). \quad (2.3)$$

Proof. Write $y = x^n$. Then $\gamma'_n = u + y'v$ and $|\gamma'_n| = \sqrt{1 + 2cy' + y'^2}$. Since $\int_0^1 y' dx = 1$,

$$2 - \text{length}(\gamma_n) = \int_0^1 (1 + y' - |u + y'v|) dx = \int_0^1 h_c(nx^{n-1}) dx.$$

Now substitute $s = nx^{n-1}$. This is an increasing bijection $(0, 1] \rightarrow (0, n]$, with $x = (s/n)^\varepsilon$ and $dx = \varepsilon (s/n)^\varepsilon ds/s$. This gives (2.2). The second expression in (2.3) comes from multiplying and dividing by $1 + s + \sqrt{1 + 2cs + s^2}$. $\square$

The graph of x^n is the case $u = (1, 0)$, $v = (0, 1)$, $c = 0$. We write

$$g(s) := h_0(s) = 1 + s - \sqrt{1 + s^2}, \quad D(n) = 2 - L(n) = \varepsilon n^{-\varepsilon} \int_0^n g(s) s^{\varepsilon-1} ds. \quad (2.4)$$

Two features of g drive everything that follows. First, $g(s) \approx s$ for small s , so gentle slopes cost little. Second, $g(s) \rightarrow 1$ as $s \rightarrow \infty$, so steep slopes cost the full horizontal distance. It is this second feature that makes $\int^n g(s) ds/s$ grow like $\ln n$. Figure 2 in Section 3 shows g together with two elementary bounds.

3 Elementary bounds

Proposition 3.1. *For every real $n \geq 3$,*

$$\frac{\ln n - 1}{n - 1} - \frac{\ln^2 n}{2(n - 1)^2} \leq D(n) \leq \frac{\ln n + 1}{n - 1}.$$

Proof. We need two bounds on g . Since $1 + s + \sqrt{1 + s^2} \geq 1 + s + \max(1, s) \geq 2 \max(1, s)$, the formula $g(s) = 2s/(1 + s + \sqrt{1 + s^2})$ gives

$$0 \leq g(s) \leq \min(s, 1). \quad (3.1)$$

Also $g(s) = 1 - (\sqrt{1 + s^2} - s) = 1 - \frac{1}{s + \sqrt{1 + s^2}}$. Hence

$$g(s) \geq 1 - \frac{1}{2s} \quad (s > 0). \quad (3.2)$$

Upper bound. By (3.1),

$$\int_0^n g(s) s^{\varepsilon-1} ds \leq \int_0^1 s^{\varepsilon} ds + \int_1^n s^{\varepsilon-1} ds = \frac{1}{1 + \varepsilon} + \frac{n^{\varepsilon} - 1}{\varepsilon}.$$

Multiplying by $\varepsilon n^{-\varepsilon}$ gives $D(n) \leq \varepsilon + (1 - n^{-\varepsilon}) \leq \varepsilon + \varepsilon \ln n$, because $1 - e^{-x} \leq x$.

Lower bound. We drop the (nonnegative) integral over $[0, 1]$ and use (3.2) on $[1, n]$:

$$D(n) \geq \varepsilon n^{-\varepsilon} \int_1^n \left(s^{\varepsilon-1} - \frac{1}{2} s^{\varepsilon-2} \right) ds = (1 - n^{-\varepsilon}) - \frac{\varepsilon n^{-\varepsilon}}{2} \cdot \frac{1 - n^{\varepsilon-1}}{1 - \varepsilon} \geq (1 - n^{-\varepsilon}) - \frac{\varepsilon}{2(1 - \varepsilon)}.$$

For $n \geq 3$ we have $\varepsilon \leq \frac{1}{2}$, so $\varepsilon/(2(1 - \varepsilon)) \leq \varepsilon$. Finally, $1 - e^{-x} \geq x - x^2/2$ for $x \geq 0$. With $x = \varepsilon \ln n$, this gives the claim. $\square$

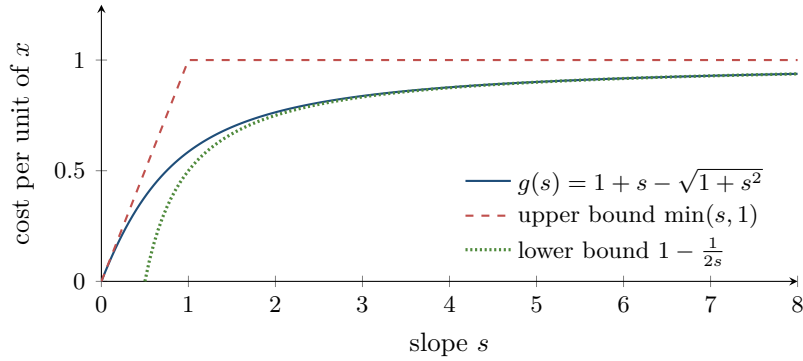


Figure 2: The defect density $g(s)$: the length lost per unit of horizontal distance along a stretch of slope s . Gentle slopes lose almost nothing; steep slopes lose almost the whole horizontal step. The bounds are (3.1) and (3.2).

Proposition 3.1 already shows that $n(2 - L(n))/\ln n \rightarrow 1$. The bounds leave room for the *constant* next to $\ln n$ anywhere in $[-1, 1]$. Section 5 shows that this constant is $1 - \ln 2 \approx 0.307$.

4 An exact formula

The function $k(s) := \sqrt{1+s^2} - s = 1 - g(s)$ is positive and decreasing, with $k(0) = 1$ and $k(s) \leq 1/(2s)$. Therefore its *Mellin transform*

$$M(\varepsilon) := \int_0^\infty k(s) s^{\varepsilon-1} ds$$

converges for $0 < \varepsilon < 1$. Mellin transforms are the natural tool for integrals like (2.4); see [3] for a survey. Here, however, we need only one explicit evaluation.

Lemma 4.1. *For $0 < \varepsilon < 1$,*

$$\varepsilon M(\varepsilon) = R(\varepsilon) = \frac{\Gamma(1 + \frac{\varepsilon}{2}) \Gamma(\frac{1-\varepsilon}{2})}{\sqrt{\pi} (1 + \varepsilon)}.$$

Proof. Put $w = k(s) = \sqrt{1+s^2} - s$. As s runs over $(0, \infty)$, w decreases from 1 to 0. Solving gives $s = (1 - w^2)/(2w)$ and $ds = -\frac{1+w^2}{2w^2} dw$. Thus

$$M(\varepsilon) = \int_0^1 w \left(\frac{1-w^2}{2w} \right)^{\varepsilon-1} \frac{1+w^2}{2w^2} dw = 2^{-\varepsilon} \int_0^1 (1-w^2)^{\varepsilon-1} w^{-\varepsilon} (1+w^2) dw.$$

Substituting $w = \sqrt{t}$ gives the Euler Beta integral $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ [2, 5.12.1]:

$$M(\varepsilon) = 2^{-\varepsilon-1} \int_0^1 (1-t)^{\varepsilon-1} (t^{-\frac{1+\varepsilon}{2}} + t^{\frac{1-\varepsilon}{2}}) dt = 2^{-\varepsilon-1} \left[B(\frac{1-\varepsilon}{2}, \varepsilon) + B(\frac{3-\varepsilon}{2}, \varepsilon) \right].$$

Write $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ and use $\Gamma(z+1) = z\Gamma(z)$, which gives $\Gamma(\frac{3+\varepsilon}{2}) = \frac{1+\varepsilon}{2}\Gamma(\frac{1+\varepsilon}{2})$. The bracket then becomes

$$\frac{\Gamma(\varepsilon)\Gamma(\frac{1-\varepsilon}{2})}{\Gamma(\frac{1+\varepsilon}{2})} \left(1 + \frac{1-\varepsilon}{1+\varepsilon} \right) = \frac{\Gamma(\varepsilon)\Gamma(\frac{1-\varepsilon}{2})}{\Gamma(\frac{3+\varepsilon}{2})}.$$

Hence $\varepsilon M(\varepsilon) = \Gamma(1+\varepsilon)\Gamma(\frac{1-\varepsilon}{2})/(2^{1+\varepsilon}\Gamma(\frac{3+\varepsilon}{2}))$. Legendre's duplication formula $\Gamma(1+\varepsilon) = \pi^{-1/2} 2^\varepsilon \Gamma(\frac{1+\varepsilon}{2})\Gamma(1+\frac{\varepsilon}{2})$ [2, 5.5.5], together with $\Gamma(\frac{3+\varepsilon}{2}) = \frac{1+\varepsilon}{2}\Gamma(\frac{1+\varepsilon}{2})$, turns this into the stated form. $\square$

Theorem 4.2. *For every real $n > 2$, with $\varepsilon = 1/(n-1)$,*

$$L(n) = 1 + \frac{R(\varepsilon)}{n^\varepsilon} - \sum_{k=1}^\infty \binom{1/2}{k} \frac{1}{n^{2k-1}((2k-1)(n-1)-1)},$$

where $\binom{1/2}{k} = \frac{(1/2)(-1/2)\cdots(3/2-k)}{k!}$, so the coefficients are $\frac{1}{2}, -\frac{1}{8}, \frac{1}{16}, -\frac{5}{128}, \dots$. The series converges absolutely.

Proof. Since $n > 2$, we have $0 < \varepsilon < 1$. By (2.4) and $g = 1 - k$,

$$\int_0^n g(s) s^{\varepsilon-1} ds = \frac{n^\varepsilon}{\varepsilon} - \int_0^\infty k(s) s^{\varepsilon-1} ds + \int_n^\infty k(s) s^{\varepsilon-1} ds.$$

Multiplying by $\varepsilon n^{-\varepsilon}$ and applying Lemma 4.1 gives

$$D(n) = 1 - \frac{R(\varepsilon)}{n^\varepsilon} + \varepsilon n^{-\varepsilon} \int_n^\infty k(s) s^{\varepsilon-1} ds.$$

For $s > 1$, the binomial series gives $k(s) = s(\sqrt{1+s^{-2}} - 1) = \sum_{k \geq 1} \binom{1/2}{k} s^{1-2k}$. Because $\sum_k |\binom{1/2}{k}| n^{-2k} < \infty$ for $n > 1$, we may integrate term by term on $[n, \infty)$ (by the Weierstrass M -test, or Tonelli's theorem). Each term gives $\int_n^\infty s^{\varepsilon-2k} ds = n^{\varepsilon+1-2k} / (2k-1-\varepsilon)$. Hence

$$\varepsilon n^{-\varepsilon} \int_n^\infty k(s) s^{\varepsilon-1} ds = \sum_{k \geq 1} \binom{1/2}{k} \frac{\varepsilon n^{1-2k}}{2k-1-\varepsilon} = \sum_{k \geq 1} \binom{1/2}{k} \frac{1}{n^{2k-1}((2k-1)(n-1)-1)}.$$

Finally, $L(n) = 2 - D(n)$. □

Remark 4.3 (The case $n = 2$). At $n = 2$ we have $\varepsilon = 1$. Both $R(\varepsilon)$ (through $\Gamma(\frac{1-\varepsilon}{2})$) and the $k = 1$ term of the series have poles, and the two poles cancel. The limit $n \rightarrow 2^+$ of the formula recovers the length of the parabola, $L(2) = \frac{\sqrt{5}}{2} + \frac{1}{4} \ln(2 + \sqrt{5})$.

Remark 4.4 (Scaled curves). The same proof applies without change to $y = cx^n$ on $[0, 1]$ for any $c > 0$ with $cn > 1$. The only difference is that the substitution now ends at $s = cn$:

$$\text{length}(cx^n) = c + \frac{R(\varepsilon)}{(cn)^\varepsilon} - \sum_{k \geq 1} \binom{1/2}{k} \frac{1}{(cn)^{2k-1}((2k-1)(n-1)-1)}.$$

Scaling shows that the length of $y = x^n$ on $[0, a]$ equals a times the length of $a^{n-1}x^n$ on $[0, 1]$, so this also covers all intervals $[0, a]$.

Remark 4.5 (Relation to the hypergeometric form). The large-argument connection formula [2, 15.8.2] expresses (1.5) as a combination of two ${}_2F_1$'s at argument $-1/n^2$. One can check that the Gamma factor multiplying $n^{-\varepsilon}$ there, $\Gamma(1 + \frac{\varepsilon}{2})\Gamma(-\frac{1+\varepsilon}{2})/\Gamma(-\frac{1}{2})$, equals $R(\varepsilon)$. What the present approach adds is an elementary route, and above all the asymptotic consequences in the next section.

5 Asymptotic expansion

We need the Taylor expansion of $\ln R$ at 0. It uses the classical values $\psi(1) = -\gamma$, $\psi(\frac{1}{2}) = -\gamma - 2\ln 2$, $\psi'(1) = \pi^2/6$, $\psi'(\frac{1}{2}) = \pi^2/2$, $\psi''(1) = -2\zeta(3)$, and $\psi''(\frac{1}{2}) = -14\zeta(3)$ [2, §5.4, §5.15], where $\psi = \Gamma'/\Gamma$.

Lemma 5.1. *As $\varepsilon \rightarrow 0$,*

$$\ln R(\varepsilon) = (\ln 2 - 1)\varepsilon + \left(\frac{\pi^2}{12} + \frac{1}{2}\right)\varepsilon^2 + \left(\frac{\zeta(3)}{4} - \frac{1}{3}\right)\varepsilon^3 + O(\varepsilon^4).$$

Proof. We have $\ln R(\varepsilon) = \ln \Gamma(1 + \frac{\varepsilon}{2}) + \ln \Gamma(\frac{1-\varepsilon}{2}) - \frac{1}{2} \ln \pi - \ln(1 + \varepsilon)$, and R is analytic near 0. The j -th Taylor coefficient of $\ln \Gamma(a + b\varepsilon)$ is $b^j \psi^{(j-1)}(a)/j!$.

- *Linear term:* $\frac{1}{2}\psi(1) - \frac{1}{2}\psi(\frac{1}{2}) - 1 = \ln 2 - 1$.
- *Quadratic term:* $\frac{1}{8}\psi'(1) + \frac{1}{8}\psi'(\frac{1}{2}) + \frac{1}{2} = \frac{\pi^2}{48} + \frac{\pi^2}{16} + \frac{1}{2}$.
- *Cubic term:* $\frac{1}{48}\psi''(1) - \frac{1}{48}\psi''(\frac{1}{2}) - \frac{1}{3} = \frac{12\zeta(3)}{48} - \frac{1}{3}$.

The γ 's cancel in the linear term. □

Corollary 5.2. *As $n \rightarrow \infty$,*

$$\begin{aligned} (a) \quad L(n) &= 2 - \frac{\ln n + 1 - \ln 2}{n - 1} + O\left(\frac{\ln^2 n}{n^2}\right), \\ (b) \quad L(n) &= 1 + \left(\frac{2}{en}\right)^{1/(n-1)} + \frac{\pi^2}{12n^2} + O\left(\frac{\ln n}{n^3}\right). \end{aligned}$$

Proof. Put $\lambda = \ln n + 1 - \ln 2$ and $b = \frac{\pi^2}{12} + \frac{1}{2}$. By Lemma 5.1,

$$\frac{R(\varepsilon)}{n^\varepsilon} = \exp(-\varepsilon\lambda + b\varepsilon^2 + O(\varepsilon^3)) = e^{-\varepsilon\lambda}(1 + b\varepsilon^2 + O(\varepsilon^3)).$$

The series in Theorem 4.2 equals $\frac{1}{2n(n-2)} + O(n^{-4})$.

For (a), note that $e^{-\varepsilon\lambda} = 1 - \varepsilon\lambda + O(\varepsilon^2 \ln^2 n)$, and every other term is $O(n^{-2})$.

For (b), note that $e^{-\varepsilon\lambda} = 1 + O(\varepsilon \ln n)$, so $e^{-\varepsilon\lambda} b\varepsilon^2 = b\varepsilon^2 + O(\ln n/n^3)$. Hence

$$L(n) = 1 + e^{-\varepsilon\lambda} + b\varepsilon^2 - \frac{1}{2n(n-2)} + O\left(\frac{\ln n}{n^3}\right).$$

Now $b\varepsilon^2 = b/n^2 + O(n^{-3})$ and $\frac{1}{2n(n-2)} = \frac{1}{2n^2} + O(n^{-3})$, and $b - \frac{1}{2} = \frac{\pi^2}{12}$. Finally, $e^{-\varepsilon\lambda} = \exp(-\frac{\ln n + 1 - \ln 2}{n-1}) = \left(\frac{2}{en}\right)^{1/(n-1)}$. □

Remark 5.3. Carrying the expansion one step further in the same way gives

$$L(n) = 1 + \left(\frac{2}{en}\right)^{\frac{1}{n-1}} + \frac{\pi^2}{12n^2} + \frac{d - b \ln n}{n^3} + O\left(\frac{\ln^2 n}{n^4}\right),$$

with $b = \frac{\pi^2}{12} + \frac{1}{2}$ and $d = b(1 + \ln 2) + \frac{\zeta(3)}{4} - \frac{4}{3} \approx 1.20631$. We have confirmed this numerically up to $n = 10^5$. All further terms can be computed the same way. They are polynomials in $\ln n$ divided by powers of n .

Figure 3 and Table 2 show the approximations at work. Figure 3(a) plots the rescaled defect $(n-1)(2-L(n))$ against $\ln n$. The points lie on the line $\ln n + 1 - \ln 2$ of Corollary 5.2(a), between the elementary bounds of Proposition 3.1. Figure 3(b) compares the errors of the two approximations in Corollary 5.2.

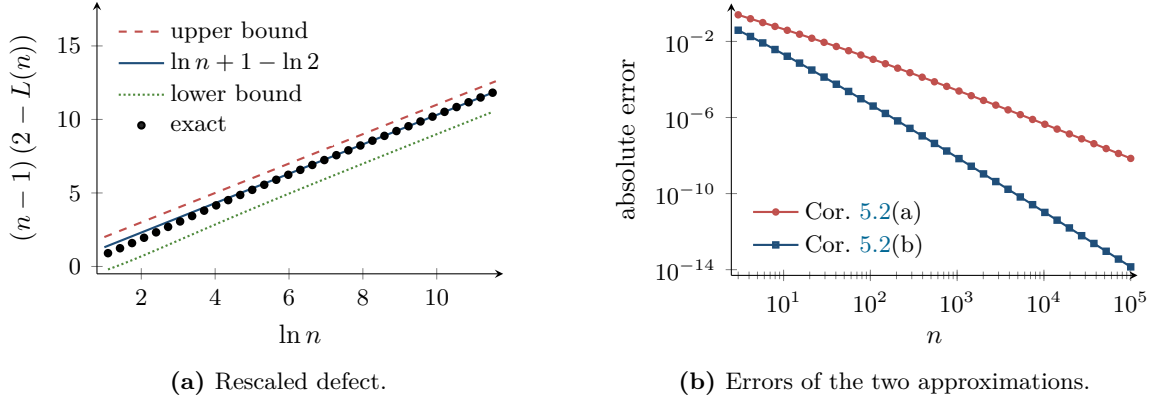


Figure 3: Accuracy of Corollary 5.2. (a) The rescaled defect (dots, from Theorem 4.2) follows the line $\ln n + 1 - \ln 2$ and stays between the bounds of Proposition 3.1. (b) The error of (a) decays like $\ln^2 n/n^2$; the error of (b) decays like $\ln n/n^3$.

Table 2: The length $L(n)$ of $y = x^n$ on $[0, 1]$, computed from Theorem 4.2 and checked against direct numerical integration for $n \leq 1000$, together with the errors of the two approximations in Corollary 5.2.

n	$L(n)$	$L(n) - \left(2 - \frac{\ln n + 1 - \ln 2}{n-1}\right)$	$L(n) - \left(1 + \left(\frac{2}{en}\right)^{\frac{1}{n-1}} + \frac{\pi^2}{12n^2}\right)$
3	1.547865654684	2.5×10^{-1}	-3.9×10^{-2}
5	1.640559757853	1.2×10^{-1}	-1.2×10^{-2}
10	1.754409376495	4.4×10^{-2}	-2.1×10^{-3}
20	1.842143162816	1.6×10^{-2}	-3.6×10^{-4}
50	1.917799927761	3.9×10^{-3}	-3.2×10^{-5}
100	1.951671753851	1.3×10^{-3}	-4.9×10^{-6}
1000	1.992804999370	2.7×10^{-5}	-7.9×10^{-9}
10000	1.999048646545	4.6×10^{-7}	-1.1×10^{-11}

6 Corners of arbitrary angle

The leading term in Corollary 5.2(a) depends only on two features of g : it is roughly linear near 0, and it tends to a limit at ∞ . The next lemma isolates this.

Lemma 6.1. *Let $h : [0, \infty) \rightarrow \mathbb{R}$ be continuous. Suppose there are constants $A, B \geq 0$ and h_∞ with $|h(s)| \leq As$ for $0 \leq s \leq 1$ and $|h(s) - h_\infty| \leq B/s$ for $s \geq 1$. Put*

$$C_h := \int_0^1 \frac{h(s)}{s} ds + \int_1^\infty \frac{h(s) - h_\infty}{s} ds.$$

Then for $n \geq 3$ and $\varepsilon = 1/(n-1)$,

$$\varepsilon n^{-\varepsilon} \int_0^n h(s) s^{\varepsilon-1} ds = \varepsilon(h_\infty \ln n + C_h) + O(\varepsilon^2 \ln^2 n),$$

where the implied constant depends only on A, B and h_∞ .

Proof. Split the integral $J = \int_0^n h(s) s^{\varepsilon-1} ds$ as

$$J = \int_0^1 h(s) s^{\varepsilon-1} ds + h_\infty \int_1^n s^{\varepsilon-1} ds + \int_1^n (h(s) - h_\infty) s^{\varepsilon-1} ds.$$

We compare each piece with its value at $\varepsilon = 0$. For $0 < s \leq 1$ we have $|s^\varepsilon - 1| \leq \varepsilon |\ln s|$, so the first piece differs from $\int_0^1 h(s) ds/s$ by at most $A\varepsilon \int_0^1 |\ln s| ds = A\varepsilon$. For $1 \leq s \leq n$ we have $|s^\varepsilon - 1| \leq \varepsilon \ln s \cdot n^\varepsilon \leq e\varepsilon \ln s$, because $n^\varepsilon = e^{\ln n/(n-1)} \leq e$. So the third piece differs from $\int_1^\infty (h - h_\infty) ds/s$ by at most $eB\varepsilon \int_1^\infty s^{-2} \ln s ds + \int_n^\infty Bs^{-2} ds \leq eB\varepsilon + B/n$. Therefore

$$J = h_\infty \frac{n^\varepsilon - 1}{\varepsilon} + C_h + E, \quad |E| \leq (A + eB + B)\varepsilon.$$

Multiplying by $\varepsilon n^{-\varepsilon}$ gives $h_\infty(1 - n^{-\varepsilon}) + \varepsilon n^{-\varepsilon}(C_h + E)$. Put $x = \varepsilon \ln n \in (0, 1]$. Then $1 - e^{-x} = x + O(x^2)$ and $e^{-x} = 1 + O(x)$, and since $\ln n \geq 1$ every error term is $O(\varepsilon^2 \ln^2 n)$. $\square$

Theorem 6.2. *Let u, v be unit vectors at angle $\theta \in [0, \pi)$, and let $L_\theta(n)$ be the length of $\gamma_n(x) = xu + x^n v$, $0 \leq x \leq 1$. Then*

$$2 - L_\theta(n) = \frac{2 \sin^2(\theta/2) \ln n + C(\theta)}{n-1} + O\left(\frac{\ln^2 n}{n^2}\right), \quad C(\theta) = 2 \sin^2 \frac{\theta}{2} + 4 \cos^2 \frac{\theta}{2} \ln \cos \frac{\theta}{2}.$$

For $\theta = \pi/2$ this gives $C = 1 - \ln 2$, in agreement with Corollary 5.2(a). For $\theta = 0$ both terms vanish, as they should, because γ_n is then a straight segment. The coefficient $2 \sin^2(\theta/2) = 1 - \cos \theta$ increases with the sharpness of the turn.

Proof. By Lemma 2.1, we must apply Lemma 6.1 to $h = h_c$ with $c = \cos \theta \in (-1, 1]$. Write $r(s) = \sqrt{1 + 2cs + s^2}$. From the second form of (2.3), we get $0 \leq h_c(s) \leq (1 - c)s$, so $A = 1 - c$. A short computation gives

$$h_c(s) - (1 - c) = \frac{(1 - c)(s - 1 - r(s))}{1 + s + r(s)}.$$

Since $|s + c| \leq r(s) \leq s + 1$, we have $|s - 1 - r(s)| \leq 2$. Therefore $h_\infty = 1 - c$ and $B = 2(1 - c)$ work.

It remains to compute C_{h_c} . Differentiation confirms that

$$G(s) = r(s) + c \ln(s + c + r(s)) - \ln \frac{1 + cs + r(s)}{s}$$

satisfies $G'(s) = r(s)/s$. (Since $r(s) \geq |s + c|$ and $r(s) \geq |1 + cs|$, with $c > -1$, both logarithms are defined for $s > 0$.) Hence $\int_\delta^N h_c(s) ds/s = [\ln s + s - G(s)]_\delta^N$, and $C_{h_c} = \lim_{\delta \rightarrow 0, N \rightarrow \infty} (\int_\delta^N h_c(s) ds/s - (1 - c) \ln N)$.

As $N \rightarrow \infty$, we have $r(N) = N + c + O(1/N)$, $s + c + r \sim 2N$, and $(1 + cN + r(N))/N \rightarrow 1 + c$. Thus

$$\ln N + N - G(N) = (1 - c) \ln N - c - c \ln 2 + \ln(1 + c) + o(1).$$

As $\delta \rightarrow 0$, we have $r(\delta) \rightarrow 1$ and $(1 + c\delta + r(\delta))/\delta \sim 2/\delta$. Thus

$$\ln \delta + \delta - G(\delta) = -1 - c \ln(1 + c) + \ln 2 + o(1).$$

Subtracting gives

$$C_{h_c} = (1 - c) + (1 + c) \ln \frac{1 + c}{2}.$$

Finally, $1 - c = 2 \sin^2 \frac{\theta}{2}$ and $\frac{1+c}{2} = \cos^2 \frac{\theta}{2}$. Lemma 6.1 states the result with $\varepsilon = 1/(n - 1)$, which is the stated form. $\square$

Figure 4(a) shows the curves γ_n for $\theta = \pi/3$. Figure 4(b) plots the two constants of Theorem 6.2 against θ , together with values obtained by computing $L_\theta(n)$ numerically at $n = 10^6$. Table 3 lists the same values. Their agreement with $C(\theta)$ is within the $O(\ln^2 n/n)$ error allowed by the theorem.

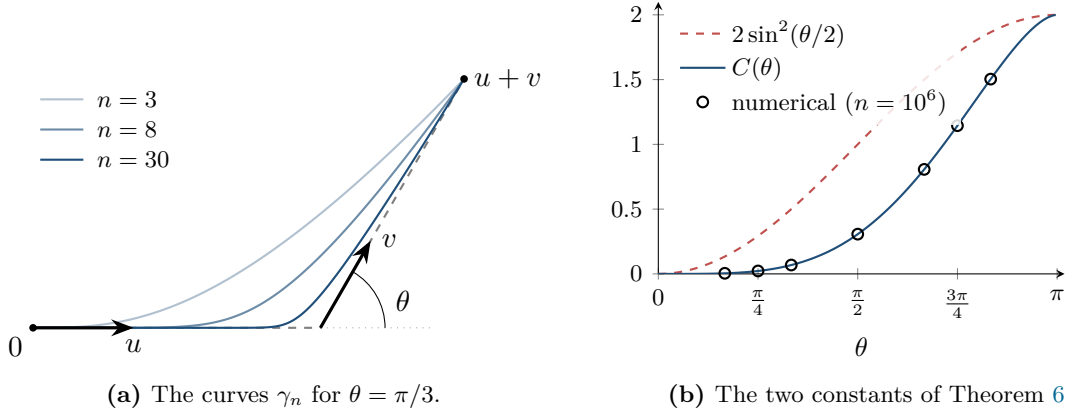


Figure 4: Corners of arbitrary angle. (a) As n grows, γ_n collapses onto the path $0 \rightarrow u \rightarrow u + v$ (dashed), which turns by the angle θ . (b) The coefficient $2 \sin^2(\theta/2)$ of $\ln n/(n - 1)$ and the constant $C(\theta)$. The circles show $(n - 1)(2 - L_\theta(n)) - 2 \sin^2(\theta/2) \ln n$ at $n = 10^6$.

Table 3: Numerical check of Theorem 6.2. The last column is computed from $L_\theta(n)$ by direct quadrature at $n = 10^6$. It should equal $C(\theta)$ up to $O(\ln^2 n/n) \approx 2 \times 10^{-4}$.

θ	$2 \sin^2(\theta/2)$	$C(\theta)$	$(n - 1)(2 - L_\theta(n)) - 2 \sin^2 \frac{\theta}{2} \ln n$
$\pi/6$	0.133975	0.0045910	0.0045781
$\pi/4$	0.292893	0.0225777	0.0225492
$\pi/3$	0.500000	0.0684769	0.0684278
$\pi/2$	1.000000	0.3068528	0.3067523
$2\pi/3$	1.500000	0.8068528	0.8066971
$3\pi/4$	1.707107	1.1444313	1.1442509
$5\pi/6$	1.866025	1.5038583	1.5036576

7 A corner without a logarithm: the squircle

Let $Q(p)$ be the length of the arc $x^p + y^p = 1$, $x, y \geq 0$, and let $P(p) = 4Q(p)$ be the full perimeter. At $p = 2$, $P = 2\pi$.

Theorem 7.1. *Let $K := 2\sqrt{2} \ln(1 + \sqrt{2}) - 2 \ln 2 = 1.1066065994\dots$. For every $p \geq 2$,*

$$\left| Q(p) - \left(2 - \frac{K}{p}\right) \right| \leq \frac{3}{p^2}, \quad \text{hence} \quad \left| P(p) - \left(8 - \frac{4K}{p}\right) \right| \leq \frac{12}{p^2}.$$

Proof. Parametrize the arc by $t \in (0, 1)$ via $x = t^{1/p}$, $y = (1 - t)^{1/p}$. Put $\varepsilon = 1/p \leq \frac{1}{2}$, $a = t^{\varepsilon-1}$ and $b = (1 - t)^{\varepsilon-1}$. Then $|x'| = a/p$ and $|y'| = b/p$, and by (2.1) (taxicab length 2),

$$2 - Q(p) = \frac{1}{p} F(\varepsilon), \quad F(\varepsilon) := \int_0^1 \Phi_\varepsilon(t) dt, \quad \Phi_\varepsilon = a + b - \sqrt{a^2 + b^2}.$$

Step 1: $F(0) = K$. At $\varepsilon = 0$, $a = 1/t$ and $b = 1/(1 - t)$. Put $q = \sqrt{t^2 + (1 - t)^2}$. Then $\Phi_0 = (1 - q)/(t(1 - t))$, and $1 - q = (1 - q^2)/(1 + q) = 2t(1 - t)/(1 + q)$. So the integrand simplifies to $\Phi_0(t) = 2/(1 + q)$. Substitute $w = 2t - 1$, so that $q^2 = (1 + w^2)/2$, and then $w = \sinh \phi$. With $A = \operatorname{arsinh} 1 = \ln(1 + \sqrt{2})$,

$$F(0) = 2 \int_0^1 \frac{dw}{1 + \sqrt{(1 + w^2)/2}} = 2\sqrt{2} \int_0^A \frac{\cosh \phi}{\sqrt{2} + \cosh \phi} d\phi = 2\sqrt{2} A - 4 \int_0^A \frac{d\phi}{\sqrt{2} + \cosh \phi}.$$

With $\tau = \tanh(\phi/2)$, the last integral becomes $\int_0^{\sqrt{2}-1} \frac{2 d\tau}{(\sqrt{2}+1)-(\sqrt{2}-1)\tau^2} = \ln \frac{\alpha+\tau}{\alpha-\tau} \Big|_{\tau=\sqrt{2}-1} = \frac{1}{2} \ln 2$, where $\alpha = \sqrt{2} + 1$. (Here we used $\tanh(A/2) = \sinh A/(1 + \cosh A) = \sqrt{2} - 1$.) Hence $F(0) = K$.

Step 2: $|F(\varepsilon) - K| \leq 3\varepsilon$. Since $\Phi_\varepsilon(t) = \Phi_\varepsilon(1 - t)$, it suffices to consider $0 < t \leq \frac{1}{2}$. Write $r = \sqrt{a^2 + b^2}$. Then $\partial\Phi/\partial a = 1 - a/r \in [0, 1)$ and likewise for b . Also $\partial_\varepsilon a = a \ln t$ and $\partial_\varepsilon b = b \ln(1 - t)$. For $t \leq \frac{1}{2}$ and $\varepsilon \leq \frac{1}{2}$ we have $1 \leq b \leq 2$ and $a \geq t^{-1/2}$. Moreover $(1 - a/r)a = \frac{ab^2}{r(r+a)} \leq \frac{b^2}{2a}$. Hence

$$|\partial_\varepsilon \Phi_\varepsilon(t)| \leq \frac{b^2}{2a} |\ln t| + b |\ln(1 - t)| \leq 2\sqrt{t} |\ln t| + 2 \ln 2 \leq \frac{4}{e} + 2 \ln 2 < 3.$$

By the mean value theorem, $|\Phi_\varepsilon(t) - \Phi_0(t)| \leq 3\varepsilon$ for every t . Integrating over $[0, 1]$ gives $|F(\varepsilon) - K| \leq 3\varepsilon$. The result follows from $Q(p) = 2 - F(1/p)/p$. $\square$

Numerical evaluation of $F'(0) = \int_0^1 \partial_\varepsilon \Phi_\varepsilon(t)|_{\varepsilon=0} dt$ gives $F'(0) = -0.5052381924\dots$. This indicates that $P(p) = 8 - 4K/p + \kappa/p^2 + o(p^{-2})$ with $\kappa = -4F'(0) = 2.0209527696\dots$, well within the bound in Theorem 7.1; see Table 4 and Figure 5(b).

Figure 5(a) shows where K comes from. Near the corner $(1, 1)$, put $x = 1 - \alpha/p$ and $y = 1 - \beta/p$. To leading order the equation $x^p + y^p = 1$ becomes $e^{-\alpha} + e^{-\beta} = 1$, a single fixed curve that does not depend on p (see Section 8).

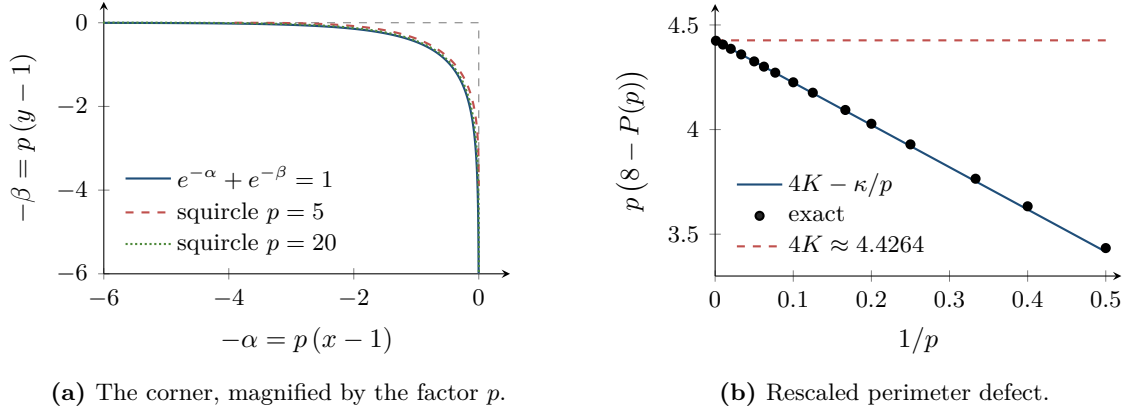


Figure 5: The squircle. (a) Near the corner $(1, 1)$, magnified by the factor p , the squircle approaches the fixed curve $e^{-\alpha} + e^{-\beta} = 1$, which meets its asymptotes (the sides of the square) exponentially fast. (b) $p(8-P(p))$ tends to $4K$ as $p \rightarrow \infty$, with slope close to $-\kappa$ (Theorem 7.1).

Table 4: Perimeter of the squircle $|x|^p + |y|^p = 1$ compared with Theorem 7.1. The last column approaches $\kappa \approx 2.02095$.

p	$P(p)$	$8 - 4K/p$	$p^2(P(p) - 8 + 4K/p)$
2	6.283185307180 ($= 2\pi$)	5.786786801	1.986
4	7.017697943564	6.893393401	1.989
10	7.577408317258	7.557357360	2.005
20	7.783710054121	7.778678680	2.013
50	7.912278464092	7.911471472	2.017
100	7.955937655831	7.955735736	2.019
1000	7.995575594378	7.995573574	2.021

8 Why the logarithm?

Formula (2.1) says that length is lost only while the curve moves in a direction strictly between horizontal and vertical. A steep piece of the curve with slope $s \gg 1$ loses the fraction $g(s) \approx 1$ of its horizontal extent. So the defect is approximately the *horizontal distance traveled while the curve is steep*.

For $y = x^n$, the slope exceeds 1 exactly when $x \geq n^{-1/(n-1)}$. This is an interval of length

$$1 - n^{-1/(n-1)} = \frac{\ln n}{n} + O\left(\frac{\ln^2 n}{n^2}\right).$$

Near $x = 1$ we have $x^n \approx e^{-n(1-x)}$. To climb from height $1/n$ (where the slope is about 1) to height 1, the curve must advance horizontally by about $\ln(n)/n$. The logarithm measures how many “ e -folds” of height the curve climbs while steep. Equivalently, x^n approaches the vertical leg $x = 1$ only logarithmically: at height y , its distance to the leg is $\ln(1/y)/n$. Figure 6 shows this for $n = 20$.

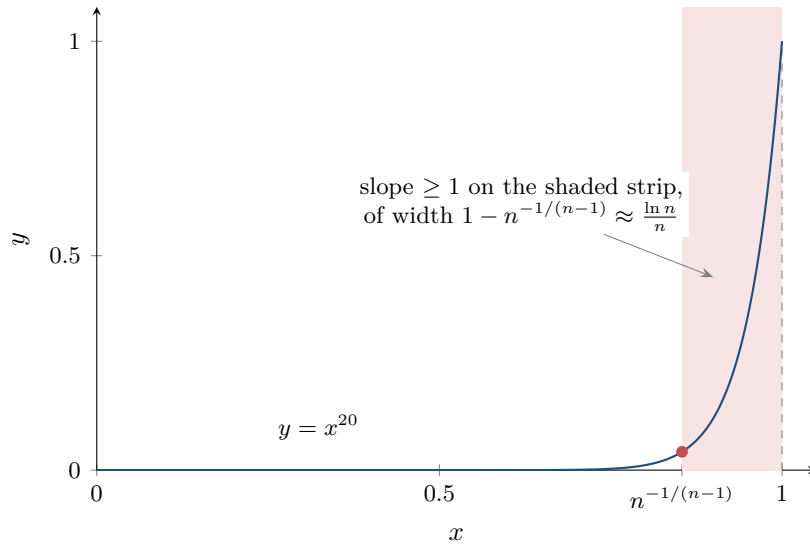


Figure 6: Where the defect of x^n comes from ($n = 20$). The curve has slope at least 1 only on the shaded strip. There it climbs from height about $1/n$ (dot) to 1, while moving horizontally by about $\ln n/n$. Almost all of that horizontal movement is lost relative to the corner path.

The squircle behaves differently. Its steep part is confined to a neighborhood of size $O(1/p)$ around the corner $(1, 1)$. Put $x = 1 - \alpha/p$ and $y = 1 - \beta/p$. The equation $x^p + y^p = 1$ becomes, to leading order, $e^{-\alpha} + e^{-\beta} = 1$: a single fixed curve, blown down by the factor $1/p$. This curve approaches its two asymptotes *exponentially* fast, so its total “diagonal travel” is finite. The constant K in Theorem 7.1 is exactly the taxicab defect of the curve $e^{-\alpha} + e^{-\beta} = 1$. (Parametrizing by $e^{-\alpha} = t$ reproduces the integral for $F(0)$.)

In short, the defect is of order $1/n$ times the logarithm of the range of heights the curve covers while steep. For the squircle this range is bounded. For x^n it is a factor of n .

9 Concluding remarks and open questions

9.1 Summary

When a family of curves collapses onto a corner, the length defect measures how much diagonal travel the curves do on the way. For the graph of x^n the defect is $(\ln n + 1 - \ln 2)/(n - 1)$ to first order. We derived this, and the complete asymptotic expansion, from an exact Gamma-function formula (Theorem 4.2). The logarithm persists for every turning angle, with coefficient $2 \sin^2(\theta/2)$ (Theorem 6.2). It is absent for the squircle, whose perimeter defect is $4K/p$ (Theorem 7.1). The difference comes down to how fast each curve approaches the sides of the corner: logarithmically for x^n , exponentially for the squircle.

9.2 Open questions

Question 1. Is there a closed form for the second squircle constant $\kappa = \lim_{p \rightarrow \infty} p^2(P(p) - 8 + 4K/p) = 2.02095276965\dots$? It equals $-4F'(0)$, where $F'(0) = \int_0^1 \partial_\varepsilon \Phi_\varepsilon|_{\varepsilon=0} dt$. This is an integral of logarithms against algebraic functions, so we expect dilogarithms. An integer-relation search against π^2 , $\ln^2 2$, $\ln^2(1 + \sqrt{2})$, $\sqrt{2} \ln 2 \ln(1 + \sqrt{2})$, $\ln 2$, and $\sqrt{2} \ln(1 + \sqrt{2})$ found no relation.

Question 2. Theorem 4.2 rests on the Mellin transform of $g = h_0$. Find the analogous exact formula for general angles, based on the Mellin transform of h_c in (2.3). We expect Legendre functions of $\cos \theta$.

Question 3. The heuristic of Section 8 suggests the following. Let $f : [0, 1] \rightarrow [0, 1]$ be smooth and increasing, with $f(0) = 0$, $f(1) = 1$ and $f'(1) > 0$. Then the graph of f^n should satisfy $2 - \text{length}(f^n) \sim \ln n / (f'(1)n)$. Prove this, and find the next term.

Question 4. What happens in higher dimensions? For example, how does the area of the graph of $z = x^n y^n$ over $[0, 1]^2$ approach its limit?

9.3 Code availability

All numerical values in this paper were computed with 30–50 significant digits using `mpmath`. The accompanying script `verify.py` recomputes them. It checks Theorem 4.2 against direct quadrature, the expansions of Corollary 5.2, the constants $C(\theta)$ and K , and Tables 2–4. The script and the L^AT_EX source of this paper are available under the MIT license at <https://github.com/zishaan1911/arc-length-of-xn>.

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